

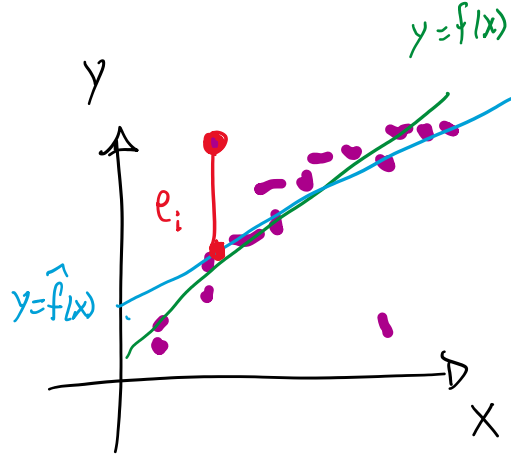
Wk3.1 Ch3 simple linear regression

Tuesday, September 22, 2026 9:31 AM

Recap:

$$Y = f(x) + \epsilon = (\beta_0 + \beta_1 X) + \epsilon$$

Assume 2 unknown parameters



Use sample (train) to find a best fit

$$\hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 X$$

Where via optimization (calc)
$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{\beta}_1 = \frac{\text{Cov}(X,Y)}{\text{Var}(X)} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

to minimize

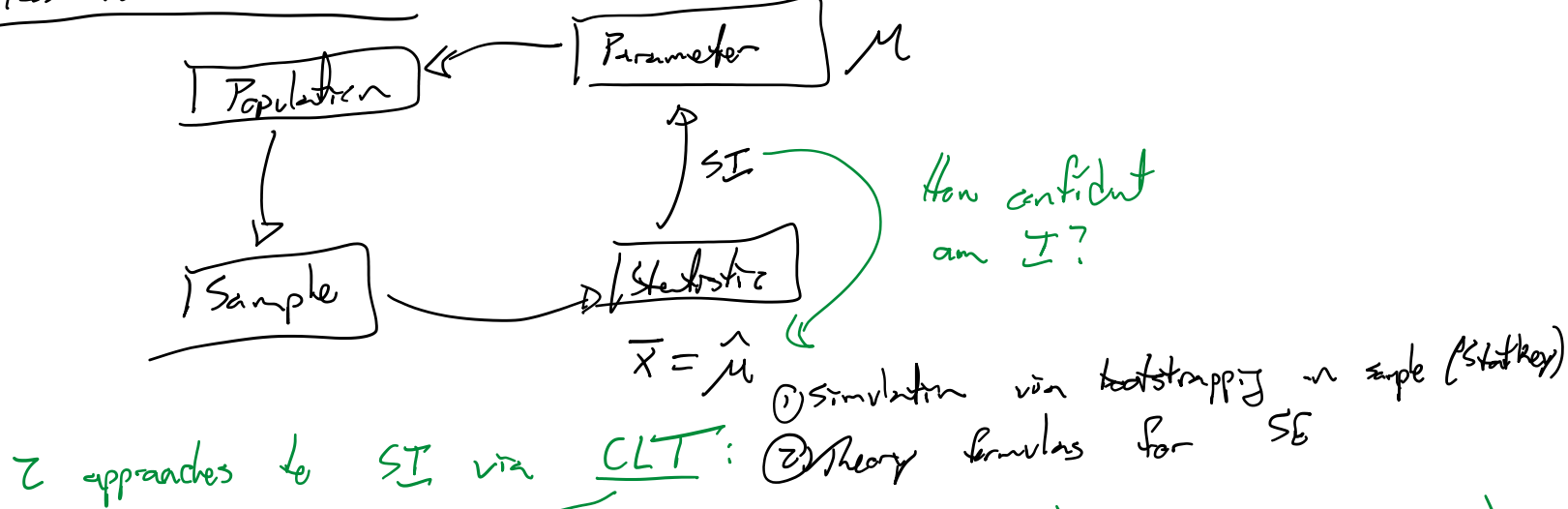
(residual sum of sqs)
$$RSS = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

How reliable are estimates $\beta_0 \approx \hat{\beta}_0$ & $\beta_1 \approx \hat{\beta}_1$? (statistical inference)

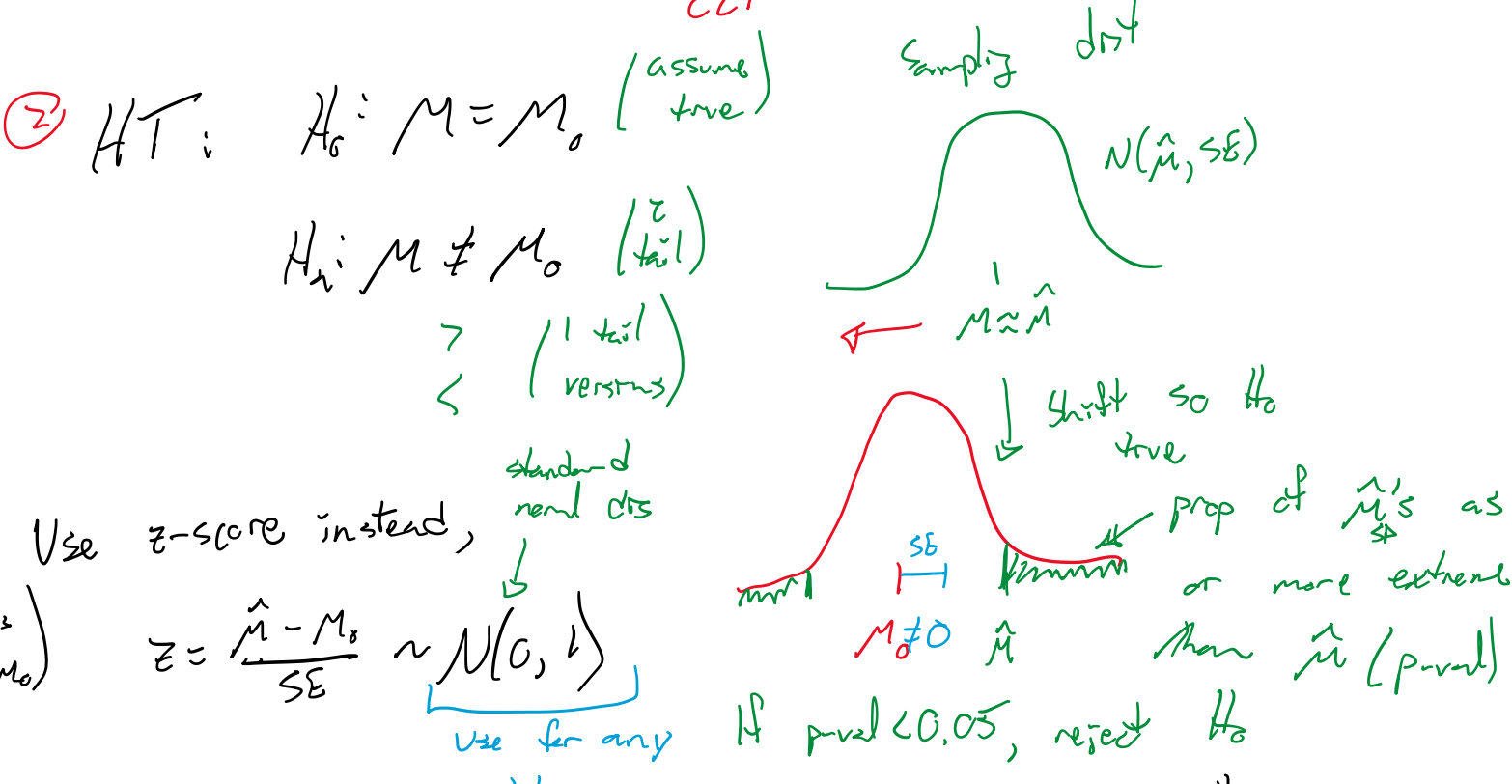
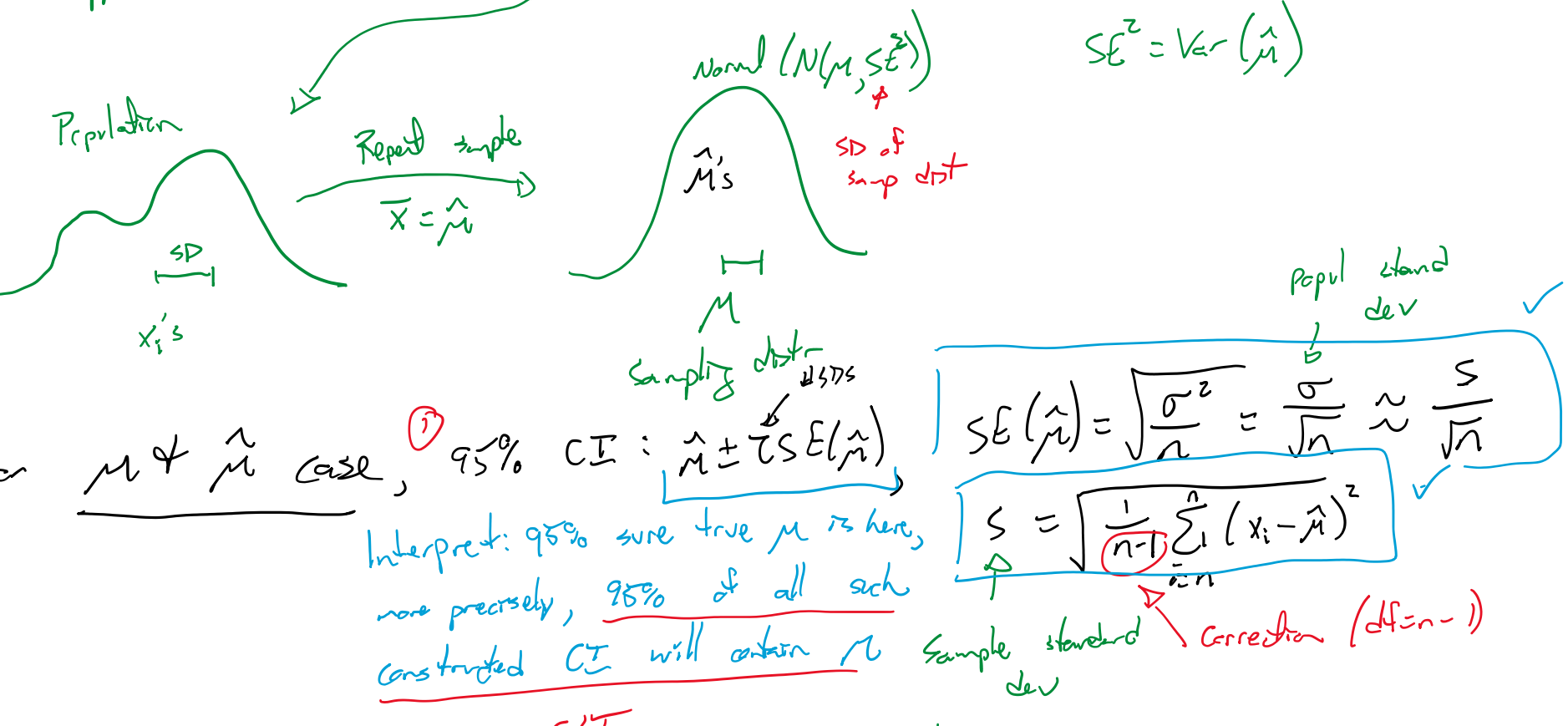
2 ways:

- ① 95% confidence interval: $\hat{\beta}_i \pm 2SE$ (sample stats)
- ② Hypothesis test: $H_0: \beta_1 = 0$ (no X-Y rel)
 $H_a: \beta_1 \neq 0$ (X-Y rel)

Fast review of SI (STAT 145)



2 approaches to SI via CLT: ② theory formulas for SE



Instead of a z-HT, often use a t-test (small samples better & same $N(0,1)$ for big samples)

(t score) $t = \frac{\hat{\mu} - \mu_0}{SE} \sim t_{df}(0,1)$ (default) (check)

Apply to sample LR: $Y = f(x) + \epsilon = (\beta_0 + \beta_1 X) + \epsilon$

(sample) $\hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 X$ (above)

95% CI: $\hat{\beta}_0 \pm 2SE(\hat{\beta}_0)$, $SE(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum (x_i - \bar{x})^2} \right)$

$\hat{\beta}_1 \pm 2SE(\hat{\beta}_1)$, $SE(\hat{\beta}_1) = \frac{\sigma^2}{\sum (x_i - \bar{x})^2}$

$$\sigma^2 = \text{Var}(\epsilon) = \frac{1}{n-2} \sum (y_i - \hat{y}_i)^2$$

(Residual SD)²

HT: $H_0: \beta_1 = 0$ (X & Y not rel)
 $H_a: \beta_1 \neq 0$

(t-score) $t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)} \sim t_{df}(0,1)$, df = n-2

Gives estimates & certainty of $\hat{\beta}_0, \hat{\beta}_1$, but what about $f(x) \approx \hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 x$ overall? 2 ways:

(old) $MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$ (y^2 units)

① $RSE = \sqrt{\frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$ (e_i stand dev, y unit)

(residual stand error)

$0 \leq R^2 \leq 1$

② R^2

$$0 \leq R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS} \leq 1$$

(Residual SS) $RSS = \sum (y_i - \hat{y}_i)^2$

(Total SS) $TSS = \sum (y_i - \bar{y})^2$ (y_i varied)

In SLR: $R^2 = r^2 = \text{Cor}(X, Y)^2$ (correlation)

$$= \left(\frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}} \right)^2$$
 (dot prod)

